

so on). This is both a form of very slow biological tapering and a form of behavioral desensitization

- Be sure to differentiate reemergence of symptoms requiring reinstitution of treatment from withdrawal symptoms
- Benzodiazepine-dependent anxiety patients and insulin-dependent diabetics are not addicted to their medications. When benzodiazepine-dependent patients stop their medication, disease symptoms can reemerge, disease symptoms can worsen (rebound), and/or withdrawal symptoms can emerge

Pharmacokinetics

- Elimination half-life 40–50 hours



Drug Interactions

- Increased depressive effects when taken with other CNS depressants (see Warnings below)



Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of

each drug to the minimum possible while still achieving therapeutic efficacy

- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur
- Dosage changes should be made in collaboration with prescriber
- Use with caution in patients with pulmonary disease; rare reports of death after initiation of benzodiazepines in patients with severe pulmonary impairment
- History of drug or alcohol abuse often creates greater risk for dependency
- Some depressed patients may experience a worsening of suicidal ideation
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)

Do Not Use

- If patient has angle-closure glaucoma
- If there is a proven allergy to clorazepate or any benzodiazepine

Special Populations

Renal Impairment

- Initial 7.5–15 mg/day in divided doses or in 1 dose at bedtime

Hepatic Impairment

- Initial 7.5–15 mg/day in divided doses or in 1 dose at bedtime

Cardiac Impairment

- Benzodiazepines have been used to treat anxiety associated with acute myocardial infarction

Elderly

- Initial 7.5–15 mg/day in divided doses or in 1 dose at bedtime



Children and Adolescents

- Not recommended for use in children under age 9
- Recommended initial dose: 7.5 mg twice a day



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Possible increased risk of birth defects when benzodiazepines are taken during pregnancy
- Because of the potential risks, clorazepate is not generally recommended as treatment for anxiety during pregnancy, especially during first trimester
- Drug should be tapered if discontinued
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy
- Seizures, even mild seizures, may cause harm to the embryo/fetus

Breast Feeding

- Some drug is found in mother's breast milk
-  Recommended either to discontinue drug or bottle feed
- Effects of benzodiazepines on nursing infants have been reported and include feeding difficulties, sedation, and weight loss

The Art of Psychopharmacology

Potential Advantages

- Rapid onset of action

Potential Disadvantages

- Euphoria may lead to abuse
- Abuse especially risky in past or present substance abusers

Primary Target Symptoms

- Panic attacks
- Anxiety
- Incidence of seizures (adjunct)

Pearls

- Can be very useful as an adjunct to SSRIs and SNRIs in the treatment of numerous anxiety disorders
- Not effective for treating psychosis as a monotherapy, but can be used as an adjunct to antipsychotics
- Not effective for treating bipolar disorder as a monotherapy, but can be used as an adjunct to mood stabilizers and antipsychotics
-  More commonly used than some other benzodiazepines for treating alcohol withdrawal
- May both cause depression and treat depression in different patients
- When using to treat insomnia, remember that insomnia may be a symptom of some other primary disorder itself, and thus warrant evaluation for comorbid psychiatric and/or medical conditions

- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment

Suggested Reading

Griffith JL , Murray GB . Clorazepate in the treatment of complex partial seizures with psychic symptomatology . *J Nerv Ment Dis* 1985 ;**173** :185 –6 .

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Kiejna A , Kantorska-Janiec M , Malyszczak K . [The use of chlorazepate dipotassium (Tranxene) in the states of restlessness and agitation] . *Psychiatr Pol* 1997 ;**31** :753 –60 .

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Mielke L , Breinbauer B , Schubert M , et al. [Comparison of the effectiveness of orally administered clorazepate dipotassium and nordiazepam on preoperative anxiety] . *Anaesthesiol Reanim* 1995 ;**20** :144 –8 .

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Rickels K , Schweizer E , Csanalosi I , Case WG , Chung H . Long-term treatment of anxiety and risk of withdrawal. Prospective comparison of clorazepate and buspirone . *Arch Gen Psychiatry* 1988 ;**45** :444 –50 .

Clozapine

Therapeutics

Brands

- Clozaril
- Leponex
- Versacloz (oral suspension)
- Fazaclo ODT (oral disintegrating tablet)
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine, serotonin, norepinephrine receptor antagonist (DSN-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotic; also a mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Treatment-resistant schizophrenia**

- **Reduction in risk of recurrent suicidal behavior in patients with schizophrenia or schizoaffective disorder**
- Treatment-resistant bipolar disorder
- Violent aggressive patients with psychosis and other brain disorders not responsive to other treatments

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
 - Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
 - Interactions at a myriad of other neurotransmitter receptors may contribute to clozapine's efficacy
- * Specifically, interactions at 5HT2C and 5HT1A receptors may contribute to efficacy for cognitive and affective symptoms in some patients
- Mechanism of efficacy for psychotic patients who do not respond to other antipsychotics is unknown but is presumed to be a mechanism other than D2 antagonism

How Long Until It Works

- Likelihood of response depends on achieving trough plasma levels of at least 350 ng/mL

- Median time to response after achieving therapeutic plasma levels (350 ng/mL) is approximately 3 weeks
- If there is no response after 3 weeks of therapeutic plasma levels, recheck plasma levels and continue titration

If It Works

- In strictly defined refractory schizophrenia, 50–60% of patients will respond to clozapine
 - The response rate to other atypical antipsychotic in the refractory patient population ranges from 0–9%
 - Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
 - Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
 - Many patients with bipolar disorder and other disorders with psychotic, aggressive, violent, impulsive, and other types of behavioral disturbances may respond to clozapine when other agents have failed
 - Perhaps 5–15% of schizophrenic patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- * Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships; superresponders are anecdotally

reported more often with clozapine than with some other antipsychotics

- Many bipolar patients may experience a reduction of symptoms by half or more
- Treatment may not only reduce mania but also prevent recurrences of mania in bipolar disorder

If It Doesn't Work

- Obtain clozapine plasma levels and continue titration
- Levels greater than 700 ng/mL are often not well tolerated
- No evidence to support dosing that results in plasma levels greater than 1000 ng/mL
- Some patients may require treatment with a conventional antipsychotic
- Some patients may require augmentation with a conventional antipsychotic or with an atypical antipsychotic (especially risperidone or amisulpride), but these are the most refractory of all psychotic patients and such treatment can be expensive
- * Consider augmentation with valproate, lamotrigine, or topiramate, but use caution with valproate as it can also cause bone marrow suppression and be sedating
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection

- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse

Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Lamotrigine
- Topiramate
- Conventional antipsychotics
- Benzodiazepines
- Lithium

Tests

- Evaluate bowel function before starting a patient on clozapine
- Lower ANC threshold for starting clozapine:
 - General population: $>1500/\mu\text{L}$
 - Benign ethnic neutropenia (BEN): $>1000/\mu\text{L}$
- Testing for myocarditis:
 - Myocarditis is rare and only occurs in the first 6 weeks of treatment
 - Baseline: check troponin I/T, C-reactive protein (CRP)

- Weekly troponin I/T and CRP for the first month
- Fever is usually benign and self-limited; suspicion of myocarditis should only be raised based on elevated troponin and other features of myocarditis
- Clozapine should be stopped if troponin $>2\times$ upper limits of normal or CRP >100 mg/L
- Cardiomyopathy is a late complication; consider annual ECG

Before starting an atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP $>140/90$ mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)

- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic
- * Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness and clouding of sensorium, even coma
- Liver function testing, ECG, general physical exam, and assessment of baseline cardiac status before starting treatment

- Liver tests may be necessary during treatment in patients who develop nausea, vomiting, or anorexia
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side effects

How Drug Causes Side Effects

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- By blocking histamine 1 receptors in the brain, it can cause sedation and possibly weight gain
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent
- Insulin regulation may be impaired by blocking pancreatic M3 muscarinic receptors

Notable Side Effects

- Orthostasis
- Sialorrhea

- Constipation
- Sedation
- Tachycardia
- Weight gain
- Dyslipidemia and hyperglycemia
- Benign fever (~20%)
- Tardive dyskinesia (reduced risk compared to other antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Severe neutropenia
- Myocarditis (only in first 6 weeks of treatment)
- Paralytic ileus
- Seizures (risk increases with dose)
- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- Pulmonary embolism (may include deep vein thrombosis or respiratory symptoms)
- Dilated cardiomyopathy

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure (more likely when clozapine is used with another agent)
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Frequent and can be significant in amount
- May increase risk for aspiration events
- Should be managed aggressively
- Can become a health problem in some
- More than for some other antipsychotics, but never say always as not a problem in everyone

Sedation



- Frequent and can be significant in amount
- Some patients may not tolerate it
- More than for some other antipsychotics, but never say always as not a problem in everyone

- Can wear off over time
- Can reemerge as dose increases and then wear off again over time

What to Do About Side Effects

- Slow titration to minimize orthostasis and sedation
 - Minimize use of other alpha 1 antagonists
 - If orthostasis remains a problem, Florines 0.1–0.3 mg qd for volume expansion (contraindicated in congestive heart failure)
 - Take at bedtime to help reduce daytime sedation
- Sialorrhea management
 - Atropine 1% drops, 1–3 drops sublingually at bedtime; can use up to 3 times per day if needed
 - Ipratropium bromide 0.06% spray, 1–3 sprays intra-orally at bedtime; can use up to 3 times per day if needed
 - Avoid use of systemic anticholinergic agents, which increase risk of ileus (benztropine, glycopyrrolate, etc.)
 - Consider novel use of botulinum toxin injections for severe cases
- Constipation management
 - Avoid psyllium as it may worsen symptoms
 - All patients should receive docusate 250 mg when starting clozapine

- If needed, add Miralax 17 g
- If docusate + Miralax are ineffective, add either bisacodyl or sennosides
- If constipation still remains a problem, prescribe lubiprostone 8–24 $\frac{1}{4}$ g twice per day
- Advise patients to contact a healthcare professional right away if they have difficulty having a bowel movement, do not have a bowel movement at least three times a week or less than their normal frequency, or are unable to pass gas
- Weight gain and metabolic effects
 - Consider prophylactic metformin; start at 500 mg for 1 week, then increase dose
 - All patients should be referred for lifestyle management and exercise
- Tachycardia
 - Atenolol 12.5 mg qd, increase to keep resting HR <100 bpm
- Chest pain during the first 6 weeks
 - Obtain workup for myocarditis
 - In the absence of elevated troponin and myocarditis symptoms, fever is usually self-limited and there is no need to stop clozapine
- Seizures

- Valproate for myoclonic or generalized seizures
- Avoid phenytoin and carbamazepine because of kinetic interactions

Best Augmenting Agents for Side Effects

- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- Depends on plasma levels; threshold for response is trough plasma level of 350 ng/mL

Dosage Forms

- Tablet 12.5 mg, 25 mg scored, 50 mg, 100 mg scored, 200 mg
- Orally disintegrating tablet 12.5 mg, 25 mg, 100 mg, 150 mg, 200 mg
- Oral suspension 50 mg/mL

How to Dose

- Initial 25 mg at night; increase by 25–50 mg/day every 48–72 hours as tolerated
- Obtain trough plasma level on 200 mg at bedtime
- Threshold for response is 350 ng/mL
- Levels greater than 700 ng/mL are often not well tolerated
- No evidence to support dosing that results in plasma levels greater than 1000 ng/mL
- Doses greater than 500 mg per day may require a split dose
- See also The Art of Switching, after Pearls



Dosing Tips

- Because of the monitoring schedule, prescriptions are generally given 1 week at a time for the first 6 months, then every 2 weeks for months 6–12, and then monthly after 12 months
- Plasma half-life suggests twice daily administration, but in practice it may be given once a day at night
- Prior to initiating treatment with clozapine, a baseline ANC must be at least 1500/ μ L for the general population and at least 1000/ μ L for patients with documented benign ethnic neutropenia (BEN)

Recommended ANC Monitoring for the General Population

ANC Level	Recommendation	ANC Monitoring
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Recommended ANC Monitoring for the General Population

ANC Level	Recommendation	ANC Monitoring
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- Normal range
 - (at least 1500/ μ L)
-

- Initiate treatment
- If treatment is interrupted for <30 days, continue monitoring as before
- If treatment is interrupted for 30 days or more, monitor as if new patient
- First 6 months: weekly
- Second 6 months: every 2 weeks
- After 1 year: every month
- Mild neutropenia
- (1000–1499/ μ L)

Continue treatment

- Confirm all initial reports of ANC <1500/ μ L with a repeat ANC measurement within 24 hours
- Monitor 3 times/week until ANC >1500/ μ L

- Once ANC $>1500/\mu\text{L}$, return to patient's last "normal range" ANC monitoring interval
- Moderate neutropenia
- (500–999/ μL)
- Interrupt treatment for suspected clozapine-induced neutropenia
- Recommend hematology consultation
- Confirm all initial reports of ANC $<1500/\hat{\text{I}}^{\frac{1}{4}}\text{ L}$ with a repeat ANC measurement within 24 hours
- Monitor ANC daily until $>1000/\mu\text{L}$ THEN
- Monitor 3 times/week until ANC $>1500/\mu\text{L}$
- Once ANC $>1500/\mu\text{L}$, check ANC weekly for 4 weeks, then return to patient's last "normal range" ANC monitoring interval
- Severe neutropenia
- ($<500/\mu\text{L}$)
- Interrupt treatment for suspected clozapine-induced neutropenia
- Recommend hematology consultation
- Do not rechallenge unless prescriber determines benefits outweigh risks
- Confirm all initial reports of ANC $<1500/\hat{\text{I}}^{\frac{1}{4}}\text{ L}$ with a repeat ANC measurement within 24 hours
- Monitor ANC daily until $>1000/\mu\text{L}$ THEN
- Monitor 3 times/week until ANC $>1500/\mu\text{L}$

- If patient is rechallenged, resume treatment as a new patient under “normal range” monitoring once ANC >1500/ μ L

Recommended ANC Monitoring for BEN Patients

ANC Level	Recommendation	ANC Monitoring
• Normal BEN range • (established ANC baseline >1000/μL)	• Obtain at least 2 baseline ANC levels before initiating treatment • If treatment is interrupted for <30 days, continue monitoring as before • If treatment is interrupted for 30 days or more, monitor as if new patient	• First 6 months: weekly • Second 6 months: every 2 weeks • After 1 year: every month

Recommended ANC Monitoring for BEN Patients

ANC Level	Recommendation	ANC Monitoring
- BEN neutropenia (500–999/μL)	- Continue treatment - Recommend hematology consultation	- Confirm all initial reports of ANC $<1500/\hat{I}\frac{1}{4}$ L with a repeat ANC measurement within 24 hours - Monitor 3 times/week until ANC $>1000/\mu$L or $>$patient's known baseline - Once ANC $>1000/\mu$L or above patient's known baseline, check ANC weekly for 4 weeks, then return to patient's last "normal BEN range" ANC monitoring interval

Recommended ANC Monitoring for BEN Patients

ANC Level	Recommendation	ANC Monitoring
• BEN severe neutropenia • (<500/μL)	• Interrupt treatment for suspected clozapine-induced neutropenia • Recommend hematology consultation • Do not rechallenge unless prescriber determines benefits outweigh risks	• Confirm all initial reports of ANC <1500/$\hat{I}$$\frac{1}{4}$ L with a repeat ANC measurement within 24 hours • Monitor ANC daily until >500/μL THEN • Monitor 3 times/week until ANC >patient's baseline • If patient is rechallenged, resume treatment as a new patient under "normal BEN range" monitoring once ANC >1000/μL or at patient's known baseline
